

Report

1

1.1

Let white Gaussian input $x[n]$. $x[n]$ is zero mean white noise with variance 1,

$$\mu_x = 0, \quad r_{xx}[m] = E\{x[n]x^*[n-m]\} = \delta[m].$$

$$y[n] = h[n] * x[n],$$

where $h[n]$ is the impulse response corresponding to

$$H(z) = \frac{1 - 0.9z^{-1} + 0.81z^{-2}}{1 - 2.76z^{-1} + 3.809z^{-2} - 2.654z^{-3} + 0.924z^{-4}}.$$

$$r_{yy}[m] = h[m] * h^*[-m] * r_{xx}[m].$$

we have $r_{xx}[m] = \delta[m]$ and coefficients are real

$$r_{yy}[m] = h[m] * h^*[-m] = \sum_{k=-\infty}^{\infty} h[k]h^*[k-m] = \sum_{k=-\infty}^{\infty} h[k]h[k-m].$$

System is causal, $h[n] = 0$ for $n < 0$, so in the code we can compute a long impulse response $h[n]$ and approximate the true autocorrelation by

$$r_{yy}[m] \approx \sum_k h[k]h[k-m].$$

$$R_{yy}(e^{j\omega}) = R_{xx}(e^{j\omega})|H(e^{j\omega})|^2.$$

$$R_{xx}(e^{j\omega}) = 1.$$

$$R_{yy}(e^{j\omega}) = |H(e^{j\omega})|^2.$$

True autocorrelation can also be written as the inverse DTFT of the true PSD:

$$r_{yy}[m] = \frac{1}{2\pi} \int_{-\pi}^{\pi} |H(e^{j\omega})|^2 e^{j\omega m} d\omega.$$

Plots are shown in the output cell.

1.2

Results and Comments are added in the output cells of notebook

1.3

From Part a, we obtained

$$r_{yy}[m] = h[m] * h^*[-m].$$

The power spectrum is the DTFT of the autocorrelation sequence, so

$$R_{yy}(e^{j\omega}) = \sum_{m=-\infty}^{\infty} r_{yy}[m]e^{-j\omega m}.$$

Substituting the expression for $r_{yy}[m]$,

$$R_{yy}(e^{j\omega}) = \mathcal{F}\{h[m] * h^*[-m]\}.$$

Using the convolution property of the DTFT,

$$R_{yy}(e^{j\omega}) = \mathcal{F}\{h[m]\}\mathcal{F}\{h^*[-m]\}.$$

$$\mathcal{F}\{h[m]\} = H(e^{j\omega}).$$

$$\mathcal{F}\{h^*[-m]\} = \sum_{m=-\infty}^{\infty} h^*[-m]e^{-j\omega m}.$$

Let $l = -m$. Then,

$$\mathcal{F}\{h^*[-m]\} = \sum_{l=-\infty}^{\infty} h^*[l]e^{j\omega l}.$$

$$H(e^{j\omega}) = \sum_{l=-\infty}^{\infty} h[l]e^{-j\omega l}.$$

$$\mathcal{F}\{h^*[-m]\} = H^*(e^{j\omega}).$$

$$R_{yy}(e^{j\omega}) = H(e^{j\omega})H^*(e^{j\omega}) = R_{yy}(e^{j\omega}) = |H(e^{j\omega})|^2$$

Since $z = e^{j\omega}$, $z^{-1} = e^{-j\omega}$.

$$H(e^{j\omega}) = \frac{1 - 0.9e^{-j\omega} + 0.81e^{-j2\omega}}{1 - 2.76e^{-j\omega} + 3.809e^{-j2\omega} - 2.654e^{-j3\omega} + 0.924e^{-j4\omega}}.$$

So the true power spectrum is

$$R_{yy}(e^{j\omega}) = \frac{|1 - 0.9e^{-j\omega} + 0.81e^{-j2\omega}|^2}{|1 - 2.76e^{-j\omega} + 3.809e^{-j2\omega} - 2.654e^{-j3\omega} + 0.924e^{-j4\omega}|^2}.$$

Plots are in the output cells of notebook

1.4

Plots and Comments are added in the output cells of notebook

1.5

$$N \in \{64, 128, 256, 512, 1024\},$$

we generate P independent realizations of the random process $y[n]$. For the p th realization, the periodogram estimate is

$$\hat{R}_{y,N}^{(p)}(e^{j\omega}) = \frac{1}{N} \left| Y_N^{(p)}(e^{j\omega}) \right|^2,$$

$$Y_N^{(p)}(e^{j\omega}) = \sum_{n=0}^{N-1} y^{(p)}[n] e^{-j\omega n}.$$

The sample mean and Variance of the periodogram

$$\bar{R}_{y,N}(e^{j\omega}) = \frac{1}{P} \sum_{p=1}^P \hat{R}_{y,N}^{(p)}(e^{j\omega}).$$

$$\widehat{\text{Var}} \left[\hat{R}_{y,N}(e^{j\omega}) \right] = \frac{1}{P-1} \sum_{p=1}^P \left| \hat{R}_{y,N}^{(p)}(e^{j\omega}) - \bar{R}_{y,N}(e^{j\omega}) \right|^2.$$

From the periodogram analysis in lecture, the mean of the periodogram is

$$E \left[\hat{R}_{y,N}(e^{j\omega}) \right] = \frac{1}{2\pi} \int_{-\pi}^{\pi} R_{yy}(e^{j\theta}) W_B(e^{j(\omega-\theta)}) d\theta,$$

where $W_B(e^{j\omega})$ is the DTFT of the Bartlett window. Therefore, for finite N , the periodogram is a biased estimate of the true power spectrum because the true spectrum is smoothed by the Bartlett window. so that means as N increases, the Bartlett window becomes narrower, But the variance dont go to zero as N increases.

$$\lim_{N \rightarrow \infty} E \left[\hat{R}_{y,N}(e^{j\omega}) \right] = R_{yy}(e^{j\omega}).$$

$$\text{Var} \left(\hat{R}_{y,N}(e^{j\omega}) \right) \approx R_{yy}^2(e^{j\omega}) \left[\left(\frac{\sin(N\omega)}{N \sin(\omega)} \right)^2 + 1 \right].$$

Therefore, even for large N , the variance remains large meaning periodogram is not a consistent estimator of the true PSD. **Plots and Comments are added in the output cells of notebook**

1.6

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2

2.1

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2.2

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2.3

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2.4

For a unit variance white input, the learned output can be viewed as

$$y_{ML}[n] \approx \sum_m h[m]x[n-m].$$

Cross correlation between $y_{ML}[n]$ and $x[n-k]$ and $x[n]$ has variance 1:

$$E\{y_{ML}[n]x^*[n-k]\} = E\left\{\left(\sum_m h[m]x[n-m]\right)x^*[n-k]\right\} = \sum_m h[m]E\{x[n-m]x^*[n-k]\}.$$

$$E\{x[n-m]x^*[n-k]\} = \delta[m-k].$$

$$E\{y_{ML}[n]x^*[n-k]\} = \sum_m h[m]\delta[m-k] = h[k].$$

The impulse response can be estimated by sample cross correlation. where $\mathcal{I}_k$ contains the valid indices where both $y_{ML}[n]$ and $x[n-k]$ are available, and $M_k = |\mathcal{I}_k|$.

$$\hat{h}_{ML}[k] = \frac{1}{M_k} \sum_{n \in \mathcal{I}_k} y_{ML}[n]x^*[n-k]$$

$$\hat{h}_{ML}[k] = \frac{1}{M_k} \sum_{n \in \mathcal{I}_k} y_{ML}[n]x[n-k].$$

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